Simi Shrivastava

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TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, scikit-learn, PyTorch, Matplotlib), SQL, R
- **Machine Learning:** Classification, Regression, Clustering, Time-series Analysis, NLP, Deep Learning, Hyperparameter Tuning, Cross-validation
- **Data Science:** Statistical Modeling, Predictive Modeling, A/B Testing, Hypothesis Testing, Feature Engineering, Dimensionality Reduction, EDA
- **Tools & Platforms:** Snowflake, AWS (EC2, S3, RDS, Lambda, ECR), Docker, Git, Tableau, Streamlit, DuckDB

WORK EXPERIENCE

Data Science Intern | Canaria, Remote, US **Jun 2025 – Aug 2025**

- Developed a salary benchmarking pipeline and anomaly detection framework in Python and SQL pulling from 20M+ records across 15+ job boards and ATS feeds, with ETL validation logic to catch data quality issues at ingestion.
- Incorporated U.S. Census, BLS, and county-level data to build cost-of-living adjustments into the pipeline, improving regional salary accuracy for clients across geographies.
- Implemented statistical data quality checks with BLS cross-validation, cutting manual review time by 60% and processing time by 70%, keeping salary data consistent across all source feeds.

Data Analyst | Cognizant Technology Solutions, India **Nov 2019 – Jul 2024**

- Modeled stock and index data in Python (Pandas, NumPy) and SQL, uncovering patterns in component changes and trading behavior that improved trend prediction accuracy by 20% for client reports.
- Translated performance and rebalancing findings into concrete portfolio recommendations delivered directly to stock exchange clients through structured presentations.
- Built an ML bug validation utility using kNN and Decision Trees for error classification, automating QA validation and saving $14K per month.
- Engineered an Excel reconciliation tool using Macros and SQL to validate financial data during a platform migration, shrinking an 8-hour manual process to under 5 minutes.
- Maintained Tableau and Power BI dashboards tracking index performance and data quality metrics, used regularly by clients and senior leadership to monitor market trends.

EDUCATION

University of Maryland, College Park, MD **Aug 2024 – May 2026**

Master of Science in Data Science | GPA: 3.86/4.0

Coursework: *Machine Learning, Algorithms for Data Science, Big Data Systems, Natural Language Processing, Cloud Computing*

CSVTU, India **Jul 2015 – May 2019**

Bachelor of Engineering in Electronics and Telecommunication | GPA: 8.53/10.00

ACADEMIC PROJECTS

TabDDPM: Synthetic Patient Data Generation for Hospital Readmission Prediction | Python, PyTorch

- Implemented TabDDPM from scratch in PyTorch on a 101,766-record hospital dataset with a 7.8:1 class imbalance. Used separate Gaussian and multinomial diffusion pipelines for mixed feature types. Hit ROC-AUC of 0.6616 — best among SMOTE and CTGAN.
- Traced low F1 on the readmitted class to a threshold problem, not a model problem. Moving the cutoff from 0.5 to 0.15 improved F1 by 5x, confirmed with t-SNE and JS divergence.

Ocular Disease Recognition using Machine Learning | Python, TensorFlow, scikit-learn

- Trained CNN models for multi-class eye disease classification on 10,000+ retinal fundus images, hitting 91% accuracy through hyperparameter tuning, evaluated with confusion matrices and ROC/AUC curves.
- Conducted EDA and hypothesis testing on patient demographics and lifestyle data, identifying correlations with disease prevalence that shaped feature selection and model design.

Product Experimentation Analytics Framework | Python, DuckDB, SQL, Tableau, Streamlit

- Designed an end-to-end A/B test framework on 541,909 ecommerce transactions with data quality validation (SRM checks, duplicate detection, leakage removal) and a modular SQL metric layer in DuckDB.
- Applied two-proportion z-tests, Welch t-tests, bootstrap CIs, and Mann-Whitney tests across segments, surfacing a statistically significant +3.85pp conversion lift (p=0.042) and +19.4% on mobile (p=0.034).